

Ex. No. 9	Use Process Explorer to identify suspicious processes
Date	

Description:

Process Explorer is a powerful Windows tool that provides detailed information about running processes, allowing users to monitor, troubleshoot, and detect suspicious activities. It can be used to detect potential malware or harmful processes by analyzing various aspects of running applications.

Here's a step-by-step guide on how to use Process Explorer to identify suspicious processes on your system:

Project Steps:

Step 1: Download and Set Up Process Explorer

1. Download Process Explorer:

- Go to the official Microsoft Sysinternals website and download Process Explorer: [Download Process Explorer](#).

2. Extract the Program:

- Once downloaded, extract the ZIP file to a folder.

3. Run Process Explorer:

- Open the folder where you extracted the files and run procexp64.exe (64-bit) or procexp.exe (32-bit) as Administrator by right-clicking and selecting "Run as Administrator."

Step 2: Familiarize Yourself with the Interface

When Process Explorer starts, you'll see an interface that lists all running processes, showing their hierarchy, memory usage, CPU usage, and much more.

1. Process Tree:

- The main window shows a tree of all running processes, with each child process listed under its parent.
- Processes are color-coded to indicate their status:
 - Pink:** Suspended processes.
 - Light Blue:** Processes running under the same user as you.
 - Dark Blue:** Services or processes running under system accounts.

- **Green:** New processes (appearing and disappearing quickly may be suspicious).
- **Red:** Processes that have just exited.

2. Columns:

- Various columns show information such as Process ID (PID), CPU usage, Memory usage, and more.

Step 3: Identify Suspicious Processes

Now that you are familiar with the interface, follow these steps to investigate potentially suspicious processes:

1. Look for Unfamiliar Processes:

- Scan the list of running processes and identify any names that you don't recognize. Some malware disguises itself with legitimate-looking names, but odd names can be a red flag.
- Legitimate processes should typically be from trusted sources (e.g., Microsoft, Intel, Adobe, etc.).

1. Verify Digital Signatures:

- Right-click any suspicious process and select Properties.
- Go to the Image tab and check if the process has a valid Digital Signature.
- Legitimate applications from trusted developers will have valid signatures. Processes without signatures or with invalid signatures are suspicious.

2. Check the Path of the Process:

- In the Properties window of the suspicious process, look under the Image tab for the Path.
- Verify if the process is located in a legitimate directory. For example, legitimate system processes should reside in C:\Windows\System32.
- If the process is running from a temporary folder, user folder, or a random directory, it might be malware.

3. Monitor CPU, Memory, and Disk Usage:

- Look at the CPU, Memory, and Disk usage columns.
- A process that is using an unusual amount of resources without explanation could be malicious. For example, high CPU usage on a process like svchost.exe can indicate malware.

5. Check Process Description and Company Name:

- Look at the Description and Company Name columns in Process Explorer.
- Legitimate processes usually provide clear descriptions and are tied to well-known companies.
- Lack of information or suspicious company names should prompt further investigation.

6. Check the Process Network Activity:

- Some malware communicates with external servers. You can inspect a process's network connections:
 - Right-click on the process and select Properties, then go to the TCP/IP tab to see network activity associated with the process.
 - Unexpected or unexplained external connections can indicate a malicious process.

Step 4: Perform a Google Search on Suspicious Processes

If you find a process that looks suspicious or unfamiliar:

1. Search online: Perform a quick Google search using the process name (e.g., processname.exe) to find out whether the process is known to be safe or malicious.
2. Check Malware Databases: Use online databases such as VirusTotal or ProcessLibrary to verify whether the process is known for being part of malware.

Step 5: Kill or Suspend Suspicious Processes

If you find a process that you believe is malicious:

1. Kill the Process:

- Right-click the suspicious process and select Kill Process to terminate it immediately.

2. Suspend the Process:

- If you are unsure and want to prevent the process from causing damage while you investigate further, right-click and select Suspend.

3. Remove the Source File:

- Go to the location of the file (using the Path from the Properties window) and delete the executable file if confirmed to be malware.
- Be cautious, as some malware might prevent you from deleting it. In such cases, you may need to boot into Safe Mode or use a specialized tool.

Step 6: Scan Your System

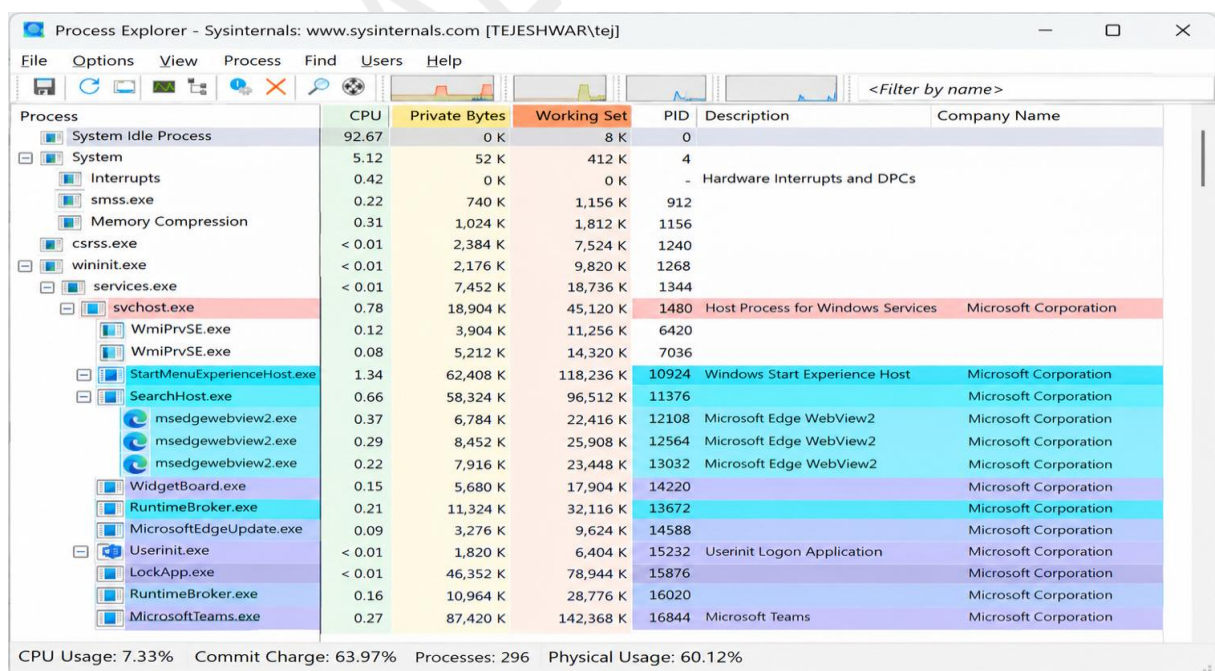
After terminating any suspicious processes, it's a good idea to:

- Run a full antivirus scan using trusted antivirus software.
- Use Malware Removal Tools like Malwarebytes or Windows Defender to perform an in-depth scan of your system.

Example of Identifying a Suspicious Process

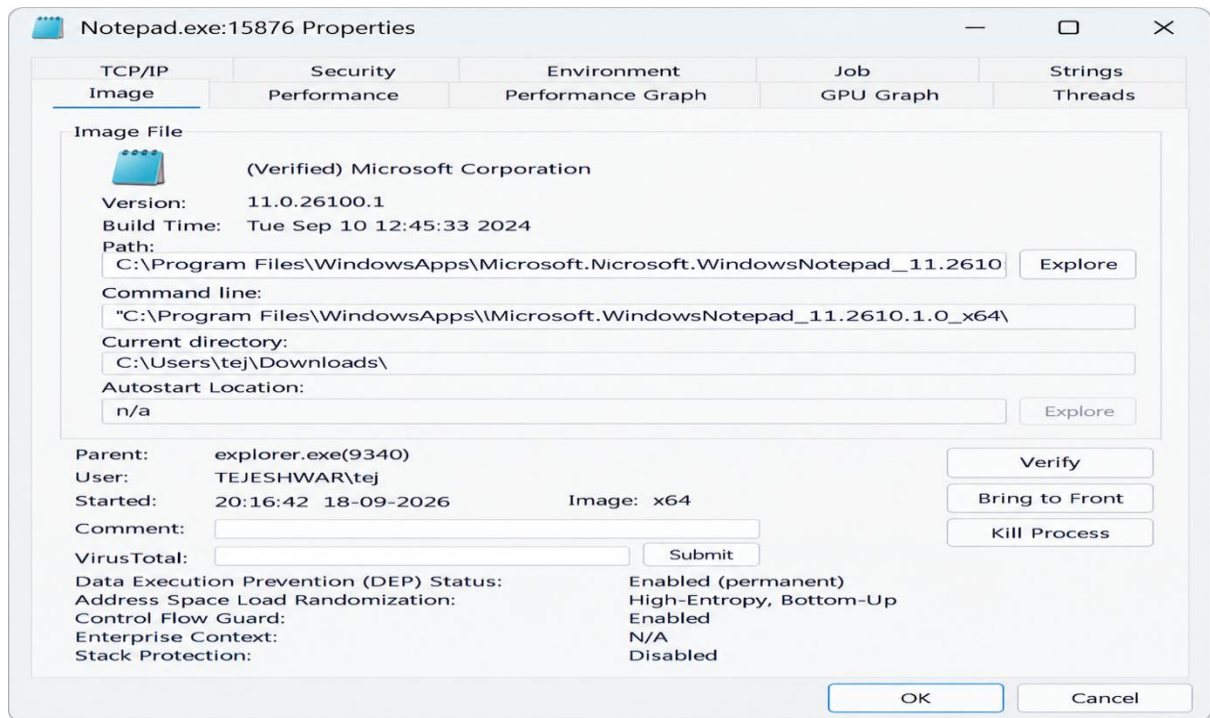
1. You notice a process called randomname123.exe using an unusually high amount of CPU.
2. Right-click it, select Properties, and find it is running from a suspicious folder, not the system folder.
3. You check the TCP/IP tab and see it is making external connections to unknown IP addresses.
4. A quick search shows that this process is often associated with a specific type of malware.
5. You kill the process and remove the executable file from the system.

Output Screenshots:



Process	CPU	Private Bytes	Working Set	PID	Description	Company Name
System Idle Process	92.67	0 K	8 K	0		
System	5.12	52 K	412 K	4		
Interrupts	0.42	0 K	0 K	-	Hardware Interrupts and DPCs	
smss.exe	0.22	740 K	1,156 K	912		
Memory Compression	0.31	1,024 K	1,812 K	1156		
csrss.exe	< 0.01	2,384 K	7,524 K	1240		
wininit.exe	< 0.01	2,176 K	9,820 K	1268		
services.exe	< 0.01	7,452 K	18,736 K	1344		
svchost.exe	0.78	18,904 K	45,120 K	1480	Host Process for Windows Services	Microsoft Corporation
WmiPrvSE.exe	0.12	3,904 K	11,256 K	6420		
WmiPrvSE.exe	0.08	5,212 K	14,320 K	7036		
StartMenuExperienceHost.exe	1.34	62,408 K	118,236 K	10924	Windows Start Experience Host	Microsoft Corporation
SearchHost.exe	0.66	58,324 K	96,512 K	11376		Microsoft Corporation
msedgewebview2.exe	0.37	6,784 K	22,416 K	12108	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe	0.29	8,452 K	25,908 K	12564	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe	0.22	7,916 K	23,448 K	13032	Microsoft Edge WebView2	Microsoft Corporation
WidgetBoard.exe	0.15	5,680 K	17,904 K	14220		Microsoft Corporation
RuntimeBroker.exe	0.21	11,324 K	32,116 K	13672		Microsoft Corporation
MicrosoftEdgeUpdate.exe	0.09	3,276 K	9,624 K	14588		Microsoft Corporation
Userinit.exe	< 0.01	1,820 K	6,404 K	15232	Userinit Logon Application	Microsoft Corporation
LockApp.exe	< 0.01	46,352 K	78,944 K	15876		Microsoft Corporation
RuntimeBroker.exe	0.16	10,964 K	28,776 K	16020		Microsoft Corporation
MicrosoftTeams.exe	0.27	87,420 K	142,368 K	16844	Microsoft Teams	Microsoft Corporation

CPU Usage: 7.33% Commit Charge: 63.97% Processes: 296 Physical Usage: 60.12%



The screenshot shows the Process Explorer window with the following columns: Process, CPU, Private Bytes, Working Set, PID, Description, and Company Name. The processes are listed in a tree view on the left, and the main table displays the following data:

Process	CPU	Private Bytes	Working Set	PID	Description	Company Name
msedge.exe		66,864 K	1,28,212 K	23488	Microsoft Edge	Microsoft Corporation
msedge.exe		9,408 K	29,248 K	18792	Microsoft Edge	Microsoft Corporation
msedge.exe	0.37	3,97,748 K	4,43,960 K	6020	Microsoft Edge	Microsoft Corporation
msedge.exe	< 0.01	81,104 K	1,40,920 K	27520	Microsoft Edge	Microsoft Corporation
msedge.exe		1,32,292 K	1,94,228 K	27568	Microsoft Edge	Microsoft Corporation
msedge.exe		8,052 K	17,952 K	26956	Microsoft Edge	Microsoft Corporation
msedge.exe		31,200 K	74,416 K	26212	Microsoft Edge	Microsoft Corporation
msedge.exe		18,308 K	34,516 K	20004	Microsoft Edge	Microsoft Corporation
MathWorksServiceHost.exe	< 0.01	78,884 K	1,77,808 K	24732	MathWorks Service Host	The MathWorks Inc.
MathWorksServiceHost-M...	< 0.01	25,652 K	60,340 K	25296	MathWorks Service Host	The MathWorks Inc.
mscopilot.exe		45,820 K	1,41,124 K	25496	Copilot	Microsoft Corporation
mscopilot.exe		2,888 K	12,928 K	25536	Copilot	Microsoft Corporation
mscopilot.exe		62,964 K	48,872 K	17956	Copilot	Microsoft Corporation
mscopilot.exe		20,380 K	47,548 K	18120	Copilot	Microsoft Corporation
mscopilot.exe		8,148 K	18,764 K	23420	Copilot	Microsoft Corporation
ONENOTEM1.exe		15,296 K	9,100 K	23460	Send to OneNote Tool	Microsoft Corporation
ONENOTEM1.exe		40,852 K	75,268 K	24332	Sysinternals Process Explorer	Sysinternals - www.sysinter...
Notepad.exe	0.12	1,21,952 K	1,85,980 K	10292		
SnippingTool.exe	1.49	54,172 K	95,112 K	3028		
BraveCrashHandler.exe		1,800 K	2,076 K	10976		
BraveCrashHandler64.exe		1,796 K	1,828 K	11004		
LenovoVantage.exe	< 0.01	1,86,252 K	26,996 K	17916	Lenovo Vantage	Lenovo
msedgewebview2.exe	< 0.01	55,394 K	11,856 K	18528	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		2,928 K	484 K	18552	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		90,012 K	1,268 K	19052	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		20,316 K	4,628 K	19064	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		8,284 K	764 K	19196	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe	< 0.01	82,468 K	18,712 K	19416	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		21,048 K	536 K	20068	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		19,988 K	212 K	20152	Microsoft Edge WebView2	Microsoft Corporation

At the bottom, the status bar shows: CPU Usage: 6.23% Commit Charge: 64.56% Processes: 296 Physical Usage: 61.21%

Process Explorer - Sysinternals: www.sysinternals.com [TEJESHWAR\dhai]

File Options View Process Find Users Help

<Filter by name>

Process	CPU	Private Bytes	Working Set	PID	Description	Company Name
msedge.exe	0.48	72,416 K	48,392 K	31216	Microsoft Edge	Microsoft Corporation
msedge.exe	1.22	85,704 K	61,208 K	18304	Microsoft Edge	Microsoft Corporation
msedge.exe	0.31	64,210 K	42,876 K	22776	Microsoft Edge	Microsoft Corporation
msedge.exe	0.12	52,318 K	36,404 K	19480	Microsoft Edge	Microsoft Corporation
msedge.exe	0.24	47,682 K	32,116 K	27632	Microsoft Edge	Microsoft Corporation
msedge.exe	0.09	39,448 K	28,572 K	24192	Microsoft Edge	Microsoft Corporation
SearchIndexerHost.exe	0.15	96,432 K	72,180 K	19356	Search Indexer Host	Microsoft Corporation
SearchIndexerService.exe	0.07	32,648 K	21,476 K	21344	Search Indexer Service	Microsoft Corporation
mscopilot.exe	0.03	8,216 K	5,420 K	24872	Copilot	Microsoft Corporation
mscopilot.exe	0.11	26,532 K	18,940 K	17644	Copilot	Microsoft Corporation
mscopilot.exe	0.18	41,360 K	29,208 K	22512	Copilot	Microsoft Corporation
mscopilot.exe	0.06	12,784 K	8,996 K	18264	Copilot	Microsoft Corporation
mscopilot.exe	0.04	10,416 K	7,312 K	24368	Copilot	Microsoft Corporation
ONENOTE.EXE	0.10	21,448 K	14,880 K	26472	Send to OneNote Tool	Microsoft Corporation
ProcessHacker64.exe	0.08	48,920 K	32,764 K	19484	Process Hacker	Process Hacker Team
Notepad.exe	0.09	1,564 K	1,276 K	27420	Notepad	Microsoft Corporation
SnippingTool.exe	1.16	62,340 K	44,816 K	31620	Snipping Tool	Microsoft Corporation
BraveCrashHandler.exe	0.05	1,724 K	1,332 K	10548		
BraveCrashHandler64.exe	0.07	2,008 K	1,612 K	11896		
LenovoVantage.exe	1.28	1,92,408 K	36,724 K	20184	Lenovo Vantage	Lenovo
msedgewebview2.exe	0.32	74,232 K	24,068 K	16852	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe	0.27	68,420 K	21,332 K	18850	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe	0.27	68,780 K	16,332 K	19560	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe	0.21	54,768 K	16,904 K	19560	Microsoft Edge WebView2	Microsoft Corporation
SDTray.exe	0.18	14,216 K	8,672 K	24208	SteelSeries Engine	SteelSeries ApS

CPU Usage: 5.84% Commit Charge: 62.17% Processes: 288 Physical Usage: 60.72%

chrome.exe:13056 Properties

Image Performance Performance Graph GPU Graph Threads TCP/IP Security Environment Job Strings

Resolve addresses

Proto.	Local Address	R..	State
UDP	dhaniya:5353	*:*	
UDP	dhaniya:5353	*:*	
UDP	dhaniya:5333	*:*	
UDP	dhaniya:5333	*:*	
UDIPv6	dhaniya:5333	*:*	
UDIPv6	dhaniya:5333	*:*	
TCP	192.168.1.24:57321	142.250.74.206:443	ESTABLISHED
TCP	192.168.1.24:57322	142.250.74.206:443	ESTABLISHED
TCP	192.168.1.24:57323	172.217.163.110:443	ESTABLISHED
TCP	192.168.1.24:57324	172.217.163.110:443	ESTABLISHED
TCP	192.168.1.24:57325	142.250.74.206:443	TIME_WAIT
TCP	192.168.1.24:57326	172.217.163.110:443	ESTABLISHED
TCP	192.168.1.24:57327	142.250.74.206:443	ESTABLISHED
TCPv6	[2401:4900:1f10:3a3::1]:57328	[2607:f8b0:4001:c1b::71]:443	ESTABLISHED
TCPv6	[2401:4900:1f10:3a3::1]:57329	[2607:f8b0:4001:c1b::71]:443	ESTABLISHED
TCPv6	[2401:4900:1f10:3a3::1]:57330	[2a00:1450:4001:829::2004]:443	TIME_WAIT

Rubrics:

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1.GitHub Activity & Submission Regularity	3	
2.Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result:

Process Explorer was successfully used to monitor running processes and identify potentially suspicious processes based on process details, paths, signatures, and resource usage. The experiment demonstrated how process behavior and network activity can be examined for forensic investigation and malware detection.